

Joseph Paul Koyi

Profile

Computer science graduate with a strong foundation in software engineering, specializing in high-performance systems. Passionate about solving complex problems programmatically and building production-grade software.

Skills

Languages	Java, C++, Python, C, SQL, NOSQL
Core Concepts	Algorithms, Data Structures, Complexity Analysis, Concurrency, OOP
Systems	Performance Optimization, Memory Management, Low-Latency Design
Tools	Git, Docker, JUnit, CMake, GitHub Actions, Linux
Practices	Full SDLC, Code Review, CI/CD, TDD

Work Experience

- Nov 2024–May 2025 **Software Engineering Intern, Ericsson**, Budapest, Hungary
- Built static analyzers for Java using SpotBugs and SEI CERT secure coding standards, achieving 99% true positive rate
 - Designed and tested custom bug detectors with high precision, validated on large-scale open-source Java projects
 - Implemented detector registration, XML configurations, and contributed to GitHub/GitLab PR workflows

Projects

NebulaPower Platform

- Built a high-throughput data processing platform in Java Spring Boot for real-time monitoring and analytics. Implemented concurrent stream processing for multiple data sources, scheduled task execution for automated compliance checks, and asynchronous event-driven alerting. Designed for performance with JPA query optimization and PostgreSQL indexing strategies.

BareMetalVM

- Engineered a low-level 16-bit virtual machine in C with custom instruction set architecture and register-based execution. Implemented memory-efficient variable-length instruction encoding, direct memory access with bounds checking, and stack-based execution model. Demonstrates deep understanding of systems programming, memory management, and performance-critical code.

Education

- Sept 2023–Feb 2026 **BSc Computer Science, Eötvös Loránd University**, Budapest, Hungary
- Relevant coursework: Algorithms & Data Structures, OOP, Operating Systems, Concurrent Programming, Computer Architecture

Certifications

- Docker Foundations Professional Certificate – Docker Inc. (Feb 2025)
- Ubuntu Linux Professional Certificate – Canonical (Feb 2025)
- Java Foundations Professional Certificate – JetBrains (Feb 2025)